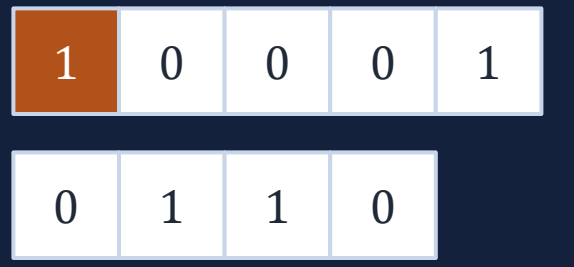
ENES 246 · DIGITAL LOGIC DESIGN

Lecture 15

Decimal Adders, Multipliers and Comparators

BCD addition and correction · the BCD adder · binary multipliers · magnitude comparators



Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapter 4, §4.6–4.8

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Today’s roadmap

1 §4.6

Decimal adder

BCD sums and the +6 fix

2 §4.7

Multiplier

AND gates and adders

3 §4.8

Comparator

Equal, greater, less

4 §4.12

Verilog

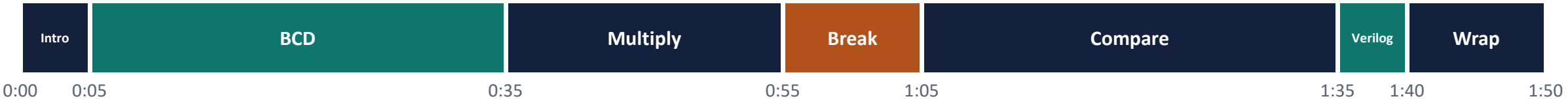
Describe them behaviorally

By the end of class you can…

- Add BCD digits and apply the +6 correction
- Draw a one-digit BCD adder and cascade digits
- Build a 2 × 2 binary multiplier and size larger ones
- Derive the magnitude comparator equations
- Design a small comparator from K-maps

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Agenda for today's session



Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome, roadmap	—
0:05 – 0:35	BCD addition and the BCD adder	Example 1 · Your turn 1
0:35 – 0:55	Binary multipliers	Example 2
0:55 – 1:05	Break	
1:05 – 1:35	Magnitude comparators	Example 3 · Your turn 2
1:35 – 1:40	Behavioral descriptions	—
1:40 – 1:50	Concept check and wrap-up	Exam 2 preview

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BCD addition

§4.6

Dec	K $Z_8Z_4Z_2Z_1$	C $S_8S_4S_2S_1$
0	0 0000	0 0000
1	0 0001	0 0001
2	0 0010	0 0010
3	0 0011	0 0011
4	0 0100	0 0100
5	0 0101	0 0101
6	0 0110	0 0110
7	0 0111	0 0111
8	0 1000	0 1000
9	0 1001	0 1001

Dec	K $Z_8Z_4Z_2Z_1$	C $S_8S_4S_2S_1$
10	0 1010	1 0000
11	0 1011	1 0001
12	0 1100	1 0010
13	0 1101	1 0011
14	0 1110	1 0100
15	0 1111	1 0101
16	1 0000	1 0110
17	1 0001	1 0111
18	1 0010	1 1000
19	1 0011	1 1001

Adding two BCD digits

Digit + digit + carry is at most $9 + 9 + 1 = 19$.

Up to 9 the binary sum is already BCD.

From 10 to 19 add 0110 (6) and send a decimal carry.

Correction needed when

$$C = K + Z_8Z_4 + Z_8Z_2$$

K covers 16–19; Z_8Z_4 and Z_8Z_2 cover 10–15.

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Example 1 — Two BCD additions

§4.6

EXAMPLE Add in BCD: (a) $4 + 5$ and (b) $8 + 9$.

4	0	1	0	0
+5	0	1	0	1
(a)	1	0	0	1

8	0	1	0	0	0
+9	0	1	0	0	1
sum	1	0	0	0	1

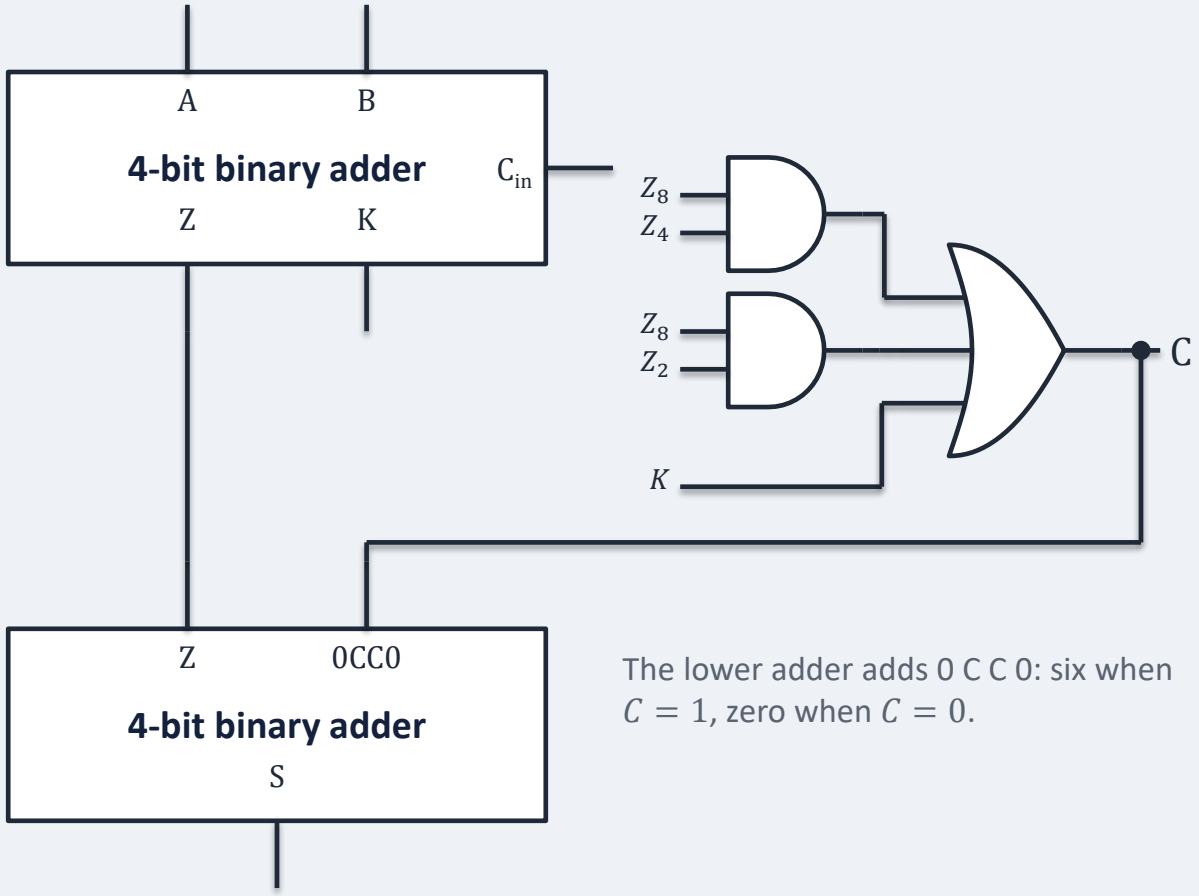
+6	0	1	1	0

1. (a) $4 = 0100$, $5 = 0101$
2. Add: $0100 + 0101 = 1001 = 9$
3. 9 is a valid digit: no correction
4. (b) $8 = 1000$, $9 = 1001$
5. Add: $1000 + 1001 = 1\,0001$ (17): $K = 1$
6. $K = 1 \rightarrow$ correct: $0001 + 0110 = 0111$
7. Decimal carry 1, digit 0111
8. **0001 0111** $= 17$ ✓

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The BCD adder

§4.6



The lower adder adds 0 C C 0: six when $C = 1$, zero when $C = 0$.

How it works

Top adder: plain binary sum of the two digits.

Logic: $C = K + Z_8Z_4 + Z_8Z_2$.

Bottom adder: adds 0 C C 0, i.e. 6 when $C = 1$ and 0 when $C = 0$.

C is the carry to the next decimal digit.

Cascade n of these for an n-digit decimal adder.

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Your turn 1 — BCD sums

§4.6

YOUR TURN

≈ 10 min

(a) Add $7 + 6$ in BCD. Is a correction needed? Which term of C fires?

(b) Add $9 + 9$ in BCD.

(c) Add $47 + 38$ with two cascaded BCD adders. Show the carry between digits.

Check every answer in decimal.

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Your turn 1 — Solution

SOLUTION

1. (a) $7 = 0111$, $6 = 0110$. Add: 1101 (13), $K = 0$
2. $Z_8Z_4 = 1 \cdot 1 = 1$, so $C = 1$: correction needed
3. $1101 + 0110 = 1\,0011 \rightarrow$ **0001 0011** $= 13$ ✓
4. (b) $1001 + 1001 = 1\,0010$: $K = 1$, so $C = 1$
5. $0010 + 0110 = 1000 \rightarrow$ **0001 1000** $= 18$ ✓
6. (c) $47 = 0100\,0111$ and $38 = 0011\,1000$
7. Units: $0111 + 1000 = 1111$ (15): $Z_8Z_4 = 1 \rightarrow C = 1$
8. $1111 + 0110 = 1\,0101$: digit **0101**, carry 1
9. Tens: $0100 + 0011 + 1 = 1000$ (8)
10. 8 is valid: $C = 0$, no correction; digit **1000**
11. $47 + 38 =$ **1000 0101** $= 85$ ✓

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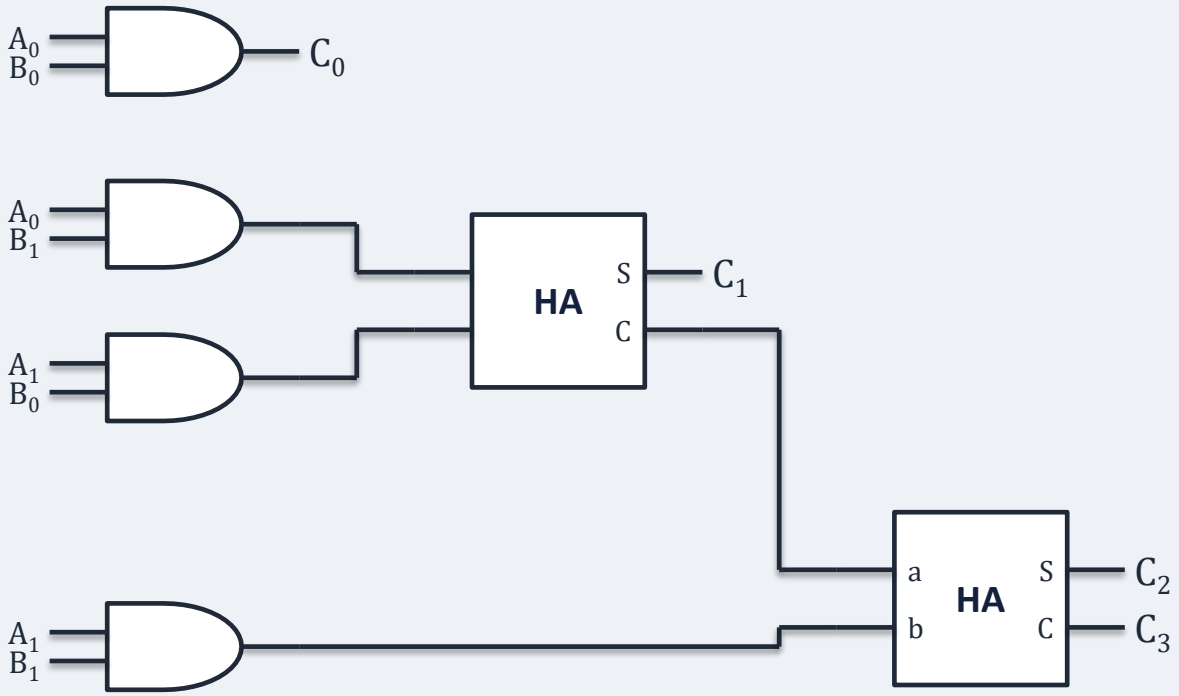
A 2 × 2 binary multiplier

Partial products

		B_1	B_0
	×	A_1	A_0
		A_0B_1	A_0B_0
+	A_1B_1	A_1B_0	
C_3	C_2	C_1	C_0

Multiplying bits is AND; adding the columns needs adders.

Circuit: 4 AND gates, 2 half adders



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Example 2 — Multiply and size

§4.7

EXAMPLE

(a) Trace 11×10 through the 2 × 2 multiplier. (b) Size a 4-bit × 3-bit multiplier.

Signal	Value	From
$C_0 = A_0B_0$	0	$1 \cdot 0$
A_0B_1, A_1B_0	1, 0	AND gates
C_1, carry	1, 0	half adder 1
A_1B_1	1	AND gate
C_2, C_3	1, 0	half adder 2

1. (a) $A = A_1A_0 = 11, B = B_1B_0 = 10$
2. $C_0 = A_0B_0 = 1 \cdot 0 = 0$
3. Partial products: $A_0B_1 = 1, A_1B_0 = 0$
4. Half adder 1: $1 + 0 \rightarrow C_1 = 1$, carry 0
5. $A_1B_1 = 1$; half adder 2: $1 + 0 \rightarrow C_2 = 1, C_3 = 0$
6. Product $C_3C_2C_1C_0 =$ **0110** $= 6 = 3 \times 2$ ✓
7. (b) One AND per pair of bits: $4 \times 3 =$ **12** ANDs
8. $K - 1 =$ **2** adders of 4 bits; product $4 + 3 =$ **7** bits

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Break

10 minutes — we resume at the 1:05 mark

Next: magnitude comparators

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The magnitude comparator

§4.8

Compare from the most significant bit

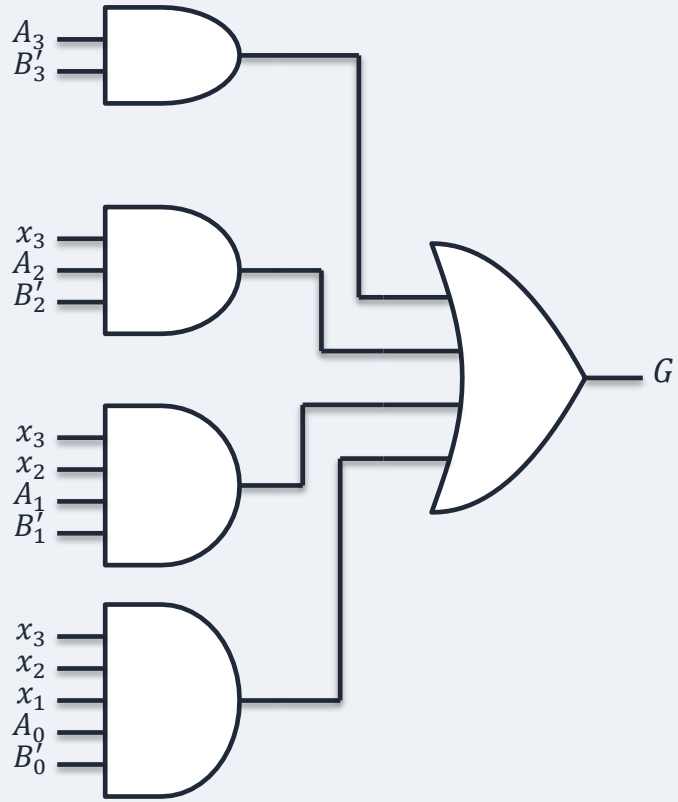
Bit i equal: $x_i = A_i B_i + A_i' B_i' = (A_i \oplus B_i)'$

$$(A = B) = x_3 x_2 x_1 x_0$$

$$(A > B) = A_3 B_3' + x_3 A_2 B_2' + x_3 x_2 A_1 B_1' + x_3 x_2 x_1 A_0 B_0'$$

$(A < B)$: swap the roles of A and B.

The (A > B) output



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Example 3 — Compare two words

§4.8

EXAMPLE Compare $A = 1011$ and $B = 1001$ with the comparator equations.

Bit i	3	2	1	0
A	1	0	1	1
B	1	0	0	1
$x_i = (A_i \oplus B_i)'$	1	1	0	1
$A_i B_i'$	0	0	1	0

1. $x_i = 1$ where the bits agree: bits 3, 2, 0 → $x = 1,1,0,1$
2. $A_i B_i' = 1$ only at bit 1 ($A_1 = 1$, $B_1 = 0$)
3. Terms of $(A > B)$: $A_3 B_3' = 0$; $x_3 A_2 B_2' = 0$
4. $x_3 x_2 A_1 B_1' = 1 \cdot 1 \cdot 1 \cdot 1 = 1$
5. So $(A > B) = 1$
6. $(A = B) = x_3 x_2 x_1 x_0 = 0$, since $x_1 = 0$
7. $11 > 9$ ✓

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Your turn 2 — A 2-bit comparator

§4.8

YOUR TURN

≈ 12 min

Design a comparator for $A = A_1A_0$ and $B = B_1B_0$ from scratch.

(a) Mark the cells of a 4-variable map ($A_1A_0B_1B_0$) where $A > B$.

(b) Find the minimum SOP for $G = (A > B)$.

(c) Write $E = (A = B)$ with XNOR gates.

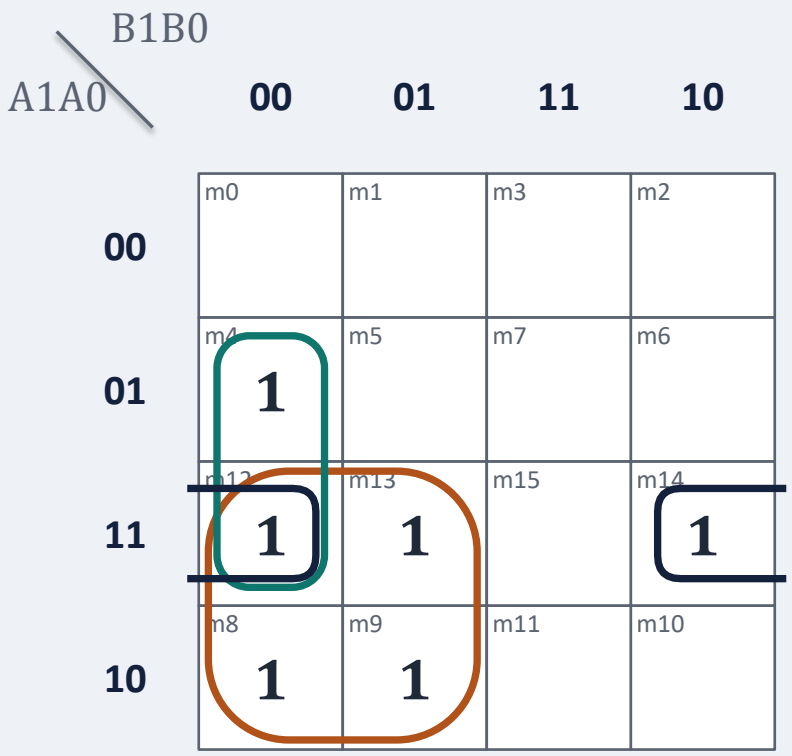
Compare your G with the general equation for $n = 2$.

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Your turn 2 — Solution

§4.8

SOLUTION



1. (a) Rows A_1A_0, columns B_1B_0. $A = 01$: only $B = 00$ → m4
2. $A = 10$: $B = 00,01$ → m8, m9; $A = 11$: m12, m13, m14
3. Quad m8, m9, m12, m13: $A_1 = 1, B_1 = 0 \rightarrow A_1B_1'$
4. Pair m4, m12: $A_0 = 1, B_1 = B_0 = 0 \rightarrow A_0B_1'B_0'$
5. Pair m12, m14: $A_1 = A_0 = 1, B_0 = 0 \rightarrow A_1A_0B_0'$
6. (b) $G = A_1B_1' + A_0B_1'B_0' + A_1A_0B_0'$
7. (c) Bit equal: $x_i = (A_i \oplus B_i)'$, an XNOR
8. $$E = x_1x_0 = (A_1 \oplus B_1)'(A_0 \oplus B_0)'$$

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Arithmetic blocks in Verilog

§4.12

```
module compare4 (input  [3:0] A, B,
                 output GT, EQ, LT);
  assign GT = (A >  B);
  assign EQ = (A == B);
  assign LT = (A <  B);
endmodule

module bcd_add (input  [3:0] A, B,
                input  Cin,
                output [3:0] S,
                output Cout);
  wire [4:0] Z = A + B + Cin;
  assign Cout = (Z > 9);
  assign S = Cout ? Z + 6 : Z;
endmodule
```

Describe the behavior

Relational operators build comparators; the synthesizer chooses the gates.

The BCD adder mirrors the hardware: binary sum, test, add 6.

Z + 6 is 5 bits; assigning it to the 4-bit S keeps the low 4 bits, which is the corrected digit.

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Concept check

1. Why is 6 added to correct a BCD sum?

2. Which sums does the term K catch, and which does $Z_8Z_4 + Z_8Z_2$?

3. How many AND gates and adders does a 5 × 4 multiplier need?

4. Why does a comparator examine the most significant bit first?

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Key takeaways

BCD correct

$$C = K + Z_8Z_4 + Z_8Z_2$$

Correction

add 0110 when $C = 1$

Multiply bits

AND

$J \times K$ array

JK ANDs, $K - 1$ adders

Equal bit

$$x_i = (A_i \oplus B_i)'$$

A > B

$$A_3B_3' + x_3A_2B_2' + \cdots$$

Every arithmetic block is gates you already know, arranged.

Homework #7 due at the start of Lecture 16

Next: review for Exam 2

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